

SW Project Documentation — Week 04

Architecture Documentation: arc42 + C4 Model + ADRs

National University of Mongolia

Summer Semester 2026

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15-Week Course

Practical Lab Exercise

Duration: 90 Min Lecture + 90 Min Lab

Date: 2026-08-30

SPRINT GOAL

WEEK 04 OBJECTIVE

Sprint Goal: Populate arc42 Sections 1 (Goals), 3 (Scope & Context), and 5 (Building Block View) in the project workspace. Embed two C4 diagrams (Context, Container) as Diagram-as-Code — following Bhatti Ch. 6 pp. 108–117 (Boxes-&Arrows, Flowchart, Swimlane). Author three Architecture Decision Records (ADRs) in MADR format, explicitly linking each back to persona pain points identified in Week 1.

PRODUCT BACKLOG (USER STORIES)

Each story follows the template "As a [role], I want [desire], so that [benefit]". Acceptance criteria are testable, with literature anchors grounding execution in assigned texts.

US-4.1 arc42 Sec 3 (Context)

"As a student, I want to populate arc42 Section 3 with external actors and neighboring systems, so that system context is immediately clear."

Acceptance Criteria:

- ≥ 3 external actors linked to Week 1 Personas.
- ≥ 2 neighboring systems with explicit data flows.
- Editable Draw.io / vector diagrams (PNG/SVG).

Literature Anchor: Bhatti et al. Ch. 6 p. 108 (Boxes & arrows) & Ch. 10 p. 169 (Organizing documentation).

US-4.2 C4 Container Diagram

"As a student, I want to draw a C4 Container diagram (max. 8 containers), so that system decomposition into deployable units is visible."

Acceptance Criteria:

- Containers have Name + Tech Stack + Responsibility.
- Follows official C4 notation conventions strictly.
- External actors from US-4.1 explicitly referenced.

Literature Anchor: Chinchilla Ch. 2 pp. 22–23; Bhatti Ch. 6 pp. 112–117 (Paper start, Reader focus, Labels).

US-4.3 3 ADRs in MADR Format

"As a student, I want to write 3 ADRs (Tech Stack, Persistence, Interface), so decisions are traceable and linked to persona pain points."

Acceptance Criteria:

- Mandatory sections: Status, Context, Decision, Consequences.
- Backward links to persona frustrations (Week 1).
- ADR Index published in Workspace 'Architecture'.

Literature Anchor: Bhatti Ch. 9 p. 149 (Docs fulfill purpose as evaluation principle for ADRs).

US-4.4 Building Block Traceability

"As a student, I want to draw the arc42 Building Block View and link components to SRS Requirement IDs to identify coverage gaps."

Acceptance Criteria:

- Each building block lists satisfied SRS Requirement IDs.
- Unmapped requirements flagged as 'orphan' for W5 planning.

Literature Anchor: arc42 Template (Sec 1, 3, 5); Chinchilla Ch. 4 p. 47 (Doc navigation & hierarchy).

DEFINITION OF DONE (DOD)

Sprint Completion Criteria

- ☐ **arc42 Sections 1, 3, and 5** fully populated and visible in workspace.
- ☐ **C4 Context & Container diagrams** stored as Diagram-as-Code (Mermaid / PlantUML) in repo.
- ☐ **Three ADRs in MADR format** published in workspace, linked to persona pain points.
- ☐ **Building Block View** contains SRS traceability links covering ≥ 80% of functional requirements.

LITERATURE-BASED MANDATORY PRACTICAL EXERCISE (UE-4)

UE-4: C4 Container Rework & Anti-Pattern Refactoring (120 Min)

Grounded in *Bhatti et al. "Docs for Developers" (Apress 2021)* and *Chinchilla "Technical Writing for Software Developers" (Packt 2024)*.

Task Instructions:

- Draw initial C4 Container diagram applying Bhatti's Three-Step Rule (Ch. 6 p. 112 "Start on paper" → p. 113 "Find starting point" → p. 116 "Use labels").
- Deliberately include at least **one architectural anti-pattern** (e.g., a "Magic Container" with vague responsibilities).
- Document rework path and refactor diagram modeling Chinchilla's "Architecture and design details" (Ch. 2 pp. 22–23).

Deliverable Submission: Before/After C4 Container diagrams + 1-page ADR justifying the refactoring step.

AGILE CEREMONIES (SPRINT EXECUTION)

Daily Standup (3 Questions)

- "How many arc42 sections grew in our workspace today?"
- "Are our C4 diagrams stored as code or static bitmaps?"
- "Which ADR decision is still waiting for team clarification?"

Sprint Retrospective (3 Questions)

- "Which arc42 section provided the highest immediate value?"
- "What level of C4 container detail did the team accept?"
- "What will we modify in our ADR format during next sprint?"

REQUIRED LITERATURE & READING LIST

Skimming with specific time allocation per chapter is part of sprint planning.

Source & Chapter	Core Focus & Pages	Relevance
Bhatti et al. Ch. 6	Visual content (pp. 101–119): Boxes-and-arrows (108), Flowcharts (110), Swimlanes (111), Drawing (112–117).	Exam High
Bhatti et al. Ch. 10	Organizing documentation (p. 169): Reader navigation and structuring.	Milestone M4
Chinchilla Ch. 2 & Ch. 4	Architecture & design details (pp. 22–23); Documentation menus & navigation (pp. 47–51).	Milestone M4
arc42 & C4 Standards	arc42 Template (Hruschka et al., Sec 1, 3, 5); C4 Model (Simon Brown - c4model.com).	Standard

Three Concluding Reflection Questions

- Which specific concept from Bhatti et al. challenged my prior documentation practice the most? *(Cite page number)*.
- Which concept from Chinchilla will I continue applying in future software projects beyond Week 16? *(Cite page number)*.
- At which point in our guiding project do I observe the largest discrepancy between "what the books advocate" and "what we actually practice"?

